# Supplementary Material for "Prospective Feedback Games for Consensus Reaching in Social Network Group Decision Making with Incomplete Preferences"

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**Abstract**

The supplement gives complete proofs of the formal results in the accompanying article—first for the general PFG framework, then for the SCMCF specialization—and documents the numerical realization of the possibly rank-deficient SCMCF membership-response QPs. It documents the full algorithmic specification with its complexity and termination analysis, and collects the experimental supplements: the mechanism illustration, the per-profile cross-mechanism endpoints, the SCMCF-ablation endpoints, and the controlled-incompleteness endpoints.

The notation and assumptions follow the main article. Section S.1 proves the general PFG results, Section S.2 verifies the SCMCF-specific properties and gives the associated QP derivations, Section S.3 documents the algorithm implementation, and Section S.4 collects the supplementary information for the experiments.

## S.1 Proofs for PFGs

This section proves the exact-potential, convex-response, compatible-response, and reference-based tradeoff results for the general PFG construction.

### S.1.1 Proof of Theorem 1

Fix DM $i$, $\boldsymbol{\xi}_{-i}$, and two feasible actions $\boldsymbol{\xi}_i, \boldsymbol{\xi}_i' \in \mathcal{X}_i$. By Definition 1, changing $\boldsymbol{\xi}_i$ leaves the prospective preference $\mathbf{p}_j^+$, and hence its loss $\ell_j$, unchanged for every $j \notin \mathcal{F}_i$. Therefore,

$$\Phi_{\boldsymbol{\vartheta}}(\boldsymbol{\xi}_i', \boldsymbol{\xi}_{-i}) - \Phi_{\boldsymbol{\vartheta}}(\boldsymbol{\xi}_i, \boldsymbol{\xi}_{-i}) = \sum_{j \in \mathcal{F}_i} \left[ \ell_j\Big(\mathbf{p}_j^+(\boldsymbol{\xi}_i', \boldsymbol{\xi}_{-i}); \boldsymbol{\vartheta}_j\Big) - \ell_j\Big(\mathbf{p}_j^+(\boldsymbol{\xi}_i, \boldsymbol{\xi}_{-i}); \boldsymbol{\vartheta}_j\Big) \right], \tag{S.1}$$

which equals $J_i(\boldsymbol{\xi}_i', \boldsymbol{\xi}_{-i}) - J_i(\boldsymbol{\xi}_i, \boldsymbol{\xi}_{-i})$ by Eq. (5). Hence, the exact-potential identity holds. If every $\mathcal{X}_i$ is compact, $\mathcal{X}$ is compact; continuity of the prospective preference maps and losses makes $\Phi_{\boldsymbol{\vartheta}}$ continuous, so it attains a global minimizer $\boldsymbol{\xi}^*$. Any unilateral action with a strictly lower local cost at $\boldsymbol{\xi}^*$ would, by the exact-potential identity, produce a strictly lower value of $\Phi_{\boldsymbol{\vartheta}}$, contradicting global minimality. Thus $\boldsymbol{\xi}^*$ is a pure-strategy Nash equilibrium. □

### S.1.2 Proof of Lemma 1

Fix $\boldsymbol{\xi}_{-i}$. For every $j \in \mathcal{F}_i$, the map $\boldsymbol{\xi}_i \mapsto \mathbf{p}_j^+(\boldsymbol{\xi}_i, \boldsymbol{\xi}_{-i})$ is affine, so its composition with the convex continuous loss $\ell_j(\cdot; \boldsymbol{\vartheta}_j)$ is convex and continuous. Their finite sum is therefore a convex continuous function on the nonempty compact convex set $\mathcal{X}_i$. It attains its minimum, the minimizer set is compact, and convexity makes that set convex. This set is exactly $\mathrm{BR}_i(\boldsymbol{\xi}_{-i})$. □

### S.1.3 Proof of Lemma 2

Let $\mathcal{U}$ have pairwise disjoint impact sets and let $\boldsymbol{\xi}'$ replace $\boldsymbol{\xi}_i$ by a feasible $\boldsymbol{\xi}'_i$ for every $i \in \mathcal{U}$. If $j \notin \bigcup_{i\in\mathcal{U}} \mathcal{F}_i$, successive use of Definition 1 shows that $\mathbf{p}_j^+$ is unchanged. If $j \in \mathcal{F}_i$ for some $i \in \mathcal{U}$, disjointness gives $j \notin \mathcal{F}_k$ for every $k \neq i$, so changing the other actions cannot alter $\mathbf{p}_j^+$. Hence the change in $\Phi_{\boldsymbol{\vartheta}}$ partitions uniquely over the sets $\mathcal{F}_i$ and equals the sum of the base-profile unilateral cost differences in Eq. (6). If every summand is negative, their sum is negative.

For distinct $i, k \in \mathcal{U}$ and every $j \in \mathcal{F}_i$, we have $j \notin \mathcal{F}_k$; therefore, changing $\boldsymbol{\xi}_k$ leaves every term of $J_i$ unchanged. Thus $J_i(\cdot, \boldsymbol{\xi}_{-i})$ and $\mathrm{BR}_i(\boldsymbol{\xi}_{-i})$ are invariant to the other updates in $\mathcal{U}$. A common deterministic selector consequently returns the same action before and after those updates, so serial execution in any order and simultaneous execution produce the same class update. □

### S.1.4 Proof of Theorem 2

For the consensus-specific loss in Eq. (7), let $\bar{\mathbf{p}}(\boldsymbol{\xi}) = [\bar{p}_\ell(\boldsymbol{\xi})]_{\ell\in\mathcal{A}}$ be the collective preference vector obtained from $\mathbf{P}^+(\boldsymbol{\xi})$, where $\bar{p}_\ell(\boldsymbol{\xi})$ is its component for alternative $\ell$. Define the auxiliary reference-drift term

$$\mathcal{D}^{\mathrm{ref}}(\boldsymbol{\xi}) := \frac{1}{N} \sum_{\ell\in\mathcal{A}} \left( \sum_j \beta_{j\ell} \right) \left[ \bar{p}_\ell(\boldsymbol{\xi}) - \bar{p}_\ell^{\mathrm{ref}} \right]^2 \geq 0. \tag{S.2}$$

Since $\sum_j \beta_{j\ell}[p_{j\ell}^+(\boldsymbol{\xi}) - \bar{p}_\ell(\boldsymbol{\xi})] = 0$, the cross term vanishes and

$$\frac{1}{N}\Phi_{\boldsymbol{\vartheta}}(\boldsymbol{\xi}) = \alpha \left[ \mathcal{V}(\mathbf{P}^+(\boldsymbol{\xi})) + \mathcal{D}^{\mathrm{ref}}(\boldsymbol{\xi}) \right] + (1-\alpha)\mathcal{M}_{\mathbf{P}}(\boldsymbol{\xi}). \tag{S.3}$$

At $\boldsymbol{\xi}^{\mathrm{ref}}$, $\mathcal{D}^{\mathrm{ref}}(\boldsymbol{\xi}^{\mathrm{ref}}) = 0$, so

$$\frac{1}{N}\Phi_{\boldsymbol{\vartheta}}(\boldsymbol{\xi}^{\mathrm{ref}}) = \alpha\mathcal{V}(\mathbf{P}^+(\boldsymbol{\xi}^{\mathrm{ref}})) + (1-\alpha)\mathcal{M}_{\mathbf{P}}(\boldsymbol{\xi}^{\mathrm{ref}}). \tag{S.4}$$

The movement term in these identities is the normalized quantity defined in Eq. (8). Using the consensus–movement trade-off objective $\mathcal{T}_{\boldsymbol{\vartheta}}$ defined in Theorem 2, if $\Phi_{\boldsymbol{\vartheta}}(\boldsymbol{\xi}) < \Phi_{\boldsymbol{\vartheta}}(\boldsymbol{\xi}^{\mathrm{ref}})$, the two identities give

$$\mathcal{T}_{\boldsymbol{\vartheta}}(\boldsymbol{\xi}) + \alpha\mathcal{D}^{\mathrm{ref}}(\boldsymbol{\xi}) < \mathcal{T}_{\boldsymbol{\vartheta}}(\boldsymbol{\xi}^{\mathrm{ref}}). \tag{S.5}$$

Because $\mathcal{D}^{\mathrm{ref}}(\boldsymbol{\xi}) \geq 0$, the strict comparison stated in Theorem 2 follows for every $\alpha \in [0, 1]$. □

## S.2 Proofs for SCMCF

This section follows the SCMCF results in the main article: locality, potential and response structure, local response computation, compatible responses, membership-induced displacement, and finite-stage consensus.

### S.2.1 Proof of Proposition 1

For fixed $\mathbf{P}$ and $\mathbf{S}_{-i}$, the prospective preference of DM $v$ is

$$\mathbf{p}_v^+(\mathbf{S}) = \mathbf{p}_v + \sum_{j\in\mathcal{N}_v^{\mathrm{out}}} \frac{\varepsilon_W + (1-2\varepsilon_W)\mathbf{s}_v^\top \mathbf{s}_j}{d_v} (\mathbf{p}_j - \mathbf{p}_v). \tag{S.6}$$

If $v = i$, subtracting Eq. (S.6) at $\mathbf{s}_i$ from its value at $\mathbf{s}'_i$ gives

$$\mathbf{p}_i^+(\mathbf{s}'_i, \mathbf{S}_{-i}) - \mathbf{p}_i^+(\mathbf{s}_i, \mathbf{S}_{-i}) = (1 - 2\varepsilon_W) \sum_{j \in \mathcal{N}_i^{\text{out}}} \frac{1}{d_i} \mathbf{s}_j^\top (\mathbf{s}'_i - \mathbf{s}_i)(\mathbf{p}_j - \mathbf{p}_i). \tag{S.7}$$

If $v \neq i$, only the summand indexed by $j = i$ can depend on $\mathbf{s}_i$. For $v \in \mathcal{N}_i^{\text{in}}$, its change is

$$\mathbf{p}_v^+(\mathbf{s}'_i, \mathbf{S}_{-i}) - \mathbf{p}_v^+(\mathbf{s}_i, \mathbf{S}_{-i}) = \frac{1 - 2\varepsilon_W}{d_v} \mathbf{s}_v^\top (\mathbf{s}'_i - \mathbf{s}_i)(\mathbf{p}_i - \mathbf{p}_v). \tag{S.8}$$

When $v \notin \{i\} \cup \mathcal{N}_i^{\text{in}}$, no summand in Eq. (S.6) depends on $\mathbf{s}_i$, so the difference is zero. These cases establish the stated piecewise identity and prove the proposition. □

### S.2.2 Proof of Corollary 1

Fix DM $i$, $\mathbf{S}_{-i}$, and two feasible membership vectors $\mathbf{s}_i, \mathbf{s}'_i \in \Delta_K$. Let $\mathbf{D}_j := \operatorname{diag}(\beta_{j1}, \ldots, \beta_{jL})$ and write $\|\mathbf{x}\|_{\mathbf{D}_j}^2 := \mathbf{x}^\top \mathbf{D}_j \mathbf{x}$. For brevity, define

$$F_j^\alpha(\mathbf{S}) := \alpha \left\| \mathbf{p}_j^+(\mathbf{S}) - \bar{\mathbf{p}}^{\text{ref}} \right\|_{\mathbf{D}_j}^2 + (1 - \alpha) \left\| \mathbf{p}_j^+(\mathbf{S}) - \mathbf{p}_j \right\|_{\mathbf{D}_j}^2. \tag{S.9}$$

Proposition 1 shows that changing $\mathbf{s}_i$ can alter only $F_j^\alpha$ for $j \in \mathcal{F}_i = \{i\} \cup \mathcal{N}_i^{\text{in}}$. Therefore,

$$\Psi(\mathbf{s}'_i, \mathbf{S}_{-i}) - \Psi(\mathbf{s}_i, \mathbf{S}_{-i}) = \sum_{j \in \mathcal{F}_i} \left[ F_j^\alpha(\mathbf{s}'_i, \mathbf{S}_{-i}) - F_j^\alpha(\mathbf{s}_i, \mathbf{S}_{-i}) \right]. \tag{S.10}$$

The right-hand side is exactly $J_i(\mathbf{s}'_i, \mathbf{S}_{-i}) - J_i(\mathbf{s}_i, \mathbf{S}_{-i})$, which proves the exact-potential identity.

The joint strategy set $\mathcal{S} = (\Delta_K)^N$ is nonempty and compact, and $\Psi$ is continuous on $\mathcal{S}$. Hence, $\Psi$ attains a global minimum at some $\widehat{\mathbf{S}}$. If this profile were not a Nash equilibrium, one DM would have a feasible unilateral deviation that strictly reduced its cost. Exactness would then give a strict reduction of $\Psi$, contradicting global minimality. Thus, $\widehat{\mathbf{S}}$ is a pure-strategy Nash equilibrium.

It remains to characterize an individual response. Fix DM $i$, $\mathbf{P}$, and $\mathbf{S}_{-i}$. The prospective preference of DM $i$ can be written as

$$\mathbf{p}_i^+(\mathbf{S}) = \mathbf{a}_i + \mathbf{B}_{ii}\mathbf{s}_i, \qquad \mathbf{a}_i := \mathbf{p}_i + \frac{\varepsilon_W}{d_i} \sum_{j \in \mathcal{N}_i^{\text{out}}} (\mathbf{p}_j - \mathbf{p}_i), \qquad \mathbf{B}_{ii} := (1 - 2\varepsilon_W) \sum_{j \in \mathcal{N}_i^{\text{out}}} \frac{1}{d_i} (\mathbf{p}_j - \mathbf{p}_i) \mathbf{s}_j^\top. \tag{S.11}$$

For every $h \in \mathcal{N}_i^{\text{in}}$, Eq. (S.18) gives $\mathbf{p}_h^+(\mathbf{S}) = \mathbf{c}_{h,-i} + \mathbf{B}_{hi}\mathbf{s}_i$, where

$$\mathbf{B}_{hi} := \frac{1 - 2\varepsilon_W}{d_h} (\mathbf{p}_i - \mathbf{p}_h) \mathbf{s}_h^\top, \qquad \mathbf{c}_{h,-i} := \mathbf{p}_h + \frac{\varepsilon_W}{d_h} \sum_{j \in \mathcal{N}_h^{\text{out}}} (\mathbf{p}_j - \mathbf{p}_h) + (1 - 2\varepsilon_W) \sum_{j \in \mathcal{N}_h^{\text{out}} \setminus \{i\}} \frac{\mathbf{s}_h^\top \mathbf{s}_j}{d_h} (\mathbf{p}_j - \mathbf{p}_h). \tag{S.12}$$

For every affected DM $j \in \mathcal{F}_i$, define the effective target $\mathbf{t}_j^\alpha := \alpha \bar{\mathbf{p}}^{\text{ref}} + (1 - \alpha)\mathbf{p}_j$. The identity

$$\alpha \|\mathbf{x} - \bar{\mathbf{p}}^{\text{ref}}\|_{\mathbf{D}_j}^2 + (1 - \alpha)\|\mathbf{x} - \mathbf{p}_j\|_{\mathbf{D}_j}^2 = \|\mathbf{x} - \mathbf{t}_j^\alpha\|_{\mathbf{D}_j}^2 + \alpha(1 - \alpha)\|\bar{\mathbf{p}}^{\text{ref}} - \mathbf{p}_j\|_{\mathbf{D}_j}^2 \tag{S.13}$$

shows that the second term on the right is independent of $\mathbf{s}_i$. Substituting the affine prospective preferences into the individual cost therefore gives

$$J_i(\mathbf{s}_i, \mathbf{S}_{-i}) = \frac{1}{2} \mathbf{s}_i^\top \mathbf{H}_i \mathbf{s}_i + \mathbf{q}_i^\top \mathbf{s}_i + \kappa_i, \tag{S.14}$$

where $\kappa_i$ is independent of $\mathbf{s}_i$ and

$$\begin{aligned}\mathbf{H}_i &:= 2\left(\mathbf{B}_{ii}^\top \mathbf{D}_i \mathbf{B}_{ii} + \sum_{h\in\mathcal{N}_i^{\text{in}}} \mathbf{B}_{hi}^\top \mathbf{D}_h \mathbf{B}_{hi}\right),\\ \mathbf{q}_i &:= 2\left[\mathbf{B}_{ii}^\top \mathbf{D}_i(\mathbf{a}_i - \mathbf{t}_i^\alpha) + \sum_{h\in\mathcal{N}_i^{\text{in}}} \mathbf{B}_{hi}^\top \mathbf{D}_h(\mathbf{c}_{h,-i} - \mathbf{t}_h^\alpha)\right].\end{aligned} \tag{S.15}$$

One admissible expression for the constant is

$$\kappa_i = \|\mathbf{a}_i - \mathbf{t}_i^\alpha\|_{\mathbf{D}_i}^2 + \sum_{h\in\mathcal{N}_i^{\text{in}}} \|\mathbf{c}_{h,-i} - \mathbf{t}_h^\alpha\|_{\mathbf{D}_h}^2 + \alpha(1-\alpha)\sum_{j\in\mathcal{F}_i} \|\bar{\mathbf{p}}^{\text{ref}} - \mathbf{p}_j\|_{\mathbf{D}_j}^2. \tag{S.16}$$

For any $\mathbf{v}\in\mathbb{R}^K$, the quadratic form associated with $\mathbf{H}_i$ is

$$\mathbf{v}^\top \mathbf{H}_i \mathbf{v} = 2\left(\|\mathbf{B}_{ii}\mathbf{v}\|_{\mathbf{D}_i}^2 + \sum_{h\in\mathcal{N}_i^{\text{in}}} \|\mathbf{B}_{hi}\mathbf{v}\|_{\mathbf{D}_h}^2\right) \geq 0.$$

Thus, $\mathbf{H}_i \succeq \mathbf{0}$ and the individual objective is convex, but $\mathbf{H}_i$ need not be positive definite. Continuity on the compact simplex guarantees a nonempty compact minimizer set, and convexity makes that set convex. Because the simplex constraints admit the relative-interior point $K^{-1}\mathbf{1}$, the KKT conditions are necessary and sufficient. Specifically, $\mathbf{s}_i^\star$ minimizes the QP evaluated at $\mathbf{S}_{-i}^\star$ if and only if there exist $\nu_i\in\mathbb{R}$ and $\boldsymbol{\lambda}_i\in\mathbb{R}_{\geq 0}^K$ such that

$$\mathbf{H}_i^\star \mathbf{s}_i^\star + \mathbf{q}_i^\star + \nu_i \mathbf{1} - \boldsymbol{\lambda}_i = \mathbf{0}, \qquad \mathbf{1}^\top \mathbf{s}_i^\star = 1, \qquad \mathbf{s}_i^\star \geq \mathbf{0}, \qquad \boldsymbol{\lambda}_i^\top \mathbf{s}_i^\star = 0. \tag{S.17}$$

These KKT conditions characterize every individual best response. Semidefiniteness permits multiple minimizers, so no inverse-Hessian or unique-response representation is available in general. □

### S.2.3 Proof of Proposition 2

The moderator broadcasts $\bar{\mathbf{p}}^{\text{ref}}$ before the response stage, while the current preference profile $\mathbf{P}$ remains fixed. Hence, every effective target $\mathbf{t}_h^\alpha = \alpha\bar{\mathbf{p}}^{\text{ref}} + (1-\alpha)\mathbf{p}_h$ is fixed in the unilateral comparison. Fix DM $i$, $\mathbf{P}$, and $\mathbf{S}_{-i}$. For each $h\in\mathcal{N}_i^{\text{in}}$, separate its prospective preference into the component independent of $\mathbf{s}_i$ and the contribution associated with DM $i$:

$$\mathbf{p}_h^+(\mathbf{S}) = \mathbf{c}_{h,-i} + \mathbf{B}_{hi}\mathbf{s}_i, \tag{S.18}$$

where

$$\mathbf{c}_{h,-i} := \mathbf{p}_h + \frac{\varepsilon_W}{d_h}\sum_{j\in\mathcal{N}_h^{\text{out}}}(\mathbf{p}_j - \mathbf{p}_h) + (1-2\varepsilon_W)\sum_{j\in\mathcal{N}_h^{\text{out}}\setminus\{i\}} \frac{\mathbf{s}_h^\top \mathbf{s}_j}{d_h}(\mathbf{p}_j - \mathbf{p}_h), \qquad \mathbf{B}_{hi} := \frac{1-2\varepsilon_W}{d_h}(\mathbf{p}_i - \mathbf{p}_h)\mathbf{s}_h^\top. \tag{S.19}$$

By Eq. (S.13), the part of DM $h$'s weighted prospective loss that depends on $\mathbf{s}_i$ is

$$\|\mathbf{c}_{h,-i} + \mathbf{B}_{hi}\mathbf{s}_i - \mathbf{t}_h^\alpha\|_{\mathbf{D}_h}^2 = \frac{1}{2}\mathbf{s}_i^\top \mathbf{H}_{hi}\mathbf{s}_i + \mathbf{q}_{hi}^\top \mathbf{s}_i + \gamma_{hi}, \tag{S.20}$$

where

$$\mathbf{H}_{hi} := 2\mathbf{B}_{hi}^{\top}\mathbf{D}_h\mathbf{B}_{hi}, \qquad \mathbf{q}_{hi} := 2\mathbf{B}_{hi}^{\top}\mathbf{D}_h(\mathbf{c}_{h,-i} - \mathbf{t}_h^{\alpha}), \qquad \gamma_{hi} := \|\mathbf{c}_{h,-i} - \mathbf{t}_h^{\alpha}\|_{\mathbf{D}_h}^2 . \tag{S.21}$$

The constants from Eqs. (S.13) and (S.20) cancel in a unilateral cost difference. Recipient $h$ therefore supplies only the pair $(\mathbf{B}_{hi}^{\top}\mathbf{D}_h\mathbf{B}_{hi}, \mathbf{B}_{hi}^{\top}\mathbf{D}_h(\mathbf{c}_{h,-i} - \mathbf{t}_h^{\alpha}))$. DM $i$ forms its analogous own-preference contribution from $\mathcal{N}_i^{\text{out}}$ and sums these quantities. This recovers $J_i(\mathbf{s}_i', \mathbf{S}_{-i}) - J_i(\mathbf{s}_i, \mathbf{S}_{-i})$ without absolute loss levels or raw preferences from the recipients' other trusted sources. □

### S.2.4 Proof of Corollary 2

By Proposition 1, SCMCF has $\mathcal{F}_i = \{i\} \cup \mathcal{N}_i^{\text{in}}$. Fix any class $\mathcal{C}_g$ in a compatible partition. For distinct $i, k \in \mathcal{C}_g$, the impact sets are disjoint by Definition 3. Lemma 2 therefore gives additive potential changes and invariant response problems within $\mathcal{C}_g$. A common deterministic selector consequently returns the same response for every member before and after the other members' updates, so serial execution in any order and simultaneous execution produce the same class update. Since $g$ is arbitrary, the conclusion holds for every class and proves Corollary 2. □

### S.2.5 Proof of Proposition 3

Let $\boldsymbol{\delta}_i = \mathbf{s}_i - \mathbf{s}_i^{\text{ref}}$, $c_{\varepsilon_W} = 1 - 2\varepsilon_W > 0$, and $\mathbf{u} = K^{-1}\mathbf{1}$. Recall the fixed-mask mean squared additional preference displacement and maximum weighted recipient exposure:

$$\begin{aligned} C_{\mathbf{\Delta}P}^{\boldsymbol{\beta}}(\mathbf{S}; \mathbf{S}^{\text{ref}}) &= \frac{1}{N}\sum_{i\in\mathcal{N}} \left\| \mathbf{p}_i^{+}(\mathbf{S}) - \mathbf{p}_i^{+}(\mathbf{S}^{\text{ref}}) \right\|_{\boldsymbol{\beta}_i}^2 , \\ \chi_{\mathbf{A}} &= \max_{i\in\mathcal{N}} \sum_{j\in\mathcal{N}_i^{\text{in}}} \frac{1}{d_j}. \end{aligned} \tag{S.22}$$

Comparing a candidate membership profile with retained memberships gives the individual next-stage displacement

$$\Delta\mathbf{p}_i^{+}(\mathbf{S}; \mathbf{S}^{\text{ref}}) := \mathbf{p}_i^{+}(\mathbf{S}) - \mathbf{p}_i^{+}(\mathbf{S}^{\text{ref}}) = \frac{c_{\varepsilon_W}}{d_i} \sum_{j\in\mathcal{N}_i^{\text{out}}} \big(\mathbf{s}_i^{\top}\mathbf{s}_j - (\mathbf{s}_i^{\text{ref}})^{\top}\mathbf{s}_j^{\text{ref}}\big)(\mathbf{p}_j - \mathbf{p}_i). \tag{S.23}$$

The co-membership-score difference is $\boldsymbol{\delta}_i^{\top}\mathbf{s}_j + (\mathbf{s}_i^{\text{ref}})^{\top}\boldsymbol{\delta}_j$. Because $\mathbf{1}^{\top}\boldsymbol{\delta}_i = \mathbf{1}^{\top}\boldsymbol{\delta}_j = 0$, it equals $\boldsymbol{\delta}_i^{\top}(\mathbf{s}_j - \mathbf{u}) + (\mathbf{s}_i^{\text{ref}} - \mathbf{u})^{\top}\boldsymbol{\delta}_j$. For every $\mathbf{s} \in \Delta_K$, $\|\mathbf{s} - \mathbf{u}\|_2^2 = \|\mathbf{s}\|_2^2 - K^{-1} \le 1 - K^{-1}$. Moreover, $\sum_{\ell}\beta_{i\ell} = 1$ and every preference component lies in $[0, 1]$, so $\|\mathbf{p}_j - \mathbf{p}_i\|_{\boldsymbol{\beta}_i} \le 1$. The Cauchy–Schwarz and triangle inequalities applied to Eq. (S.23) therefore yield

$$\left\|\Delta\mathbf{p}_i^{+}(\mathbf{S}; \mathbf{S}^{\text{ref}})\right\|_{\boldsymbol{\beta}_i} \le \frac{c_{\varepsilon_W}\sqrt{1 - K^{-1}}}{d_i} \sum_{j\in\mathcal{N}_i^{\text{out}}} \big(\|\boldsymbol{\delta}_i\|_2 + \|\boldsymbol{\delta}_j\|_2\big). \tag{S.24}$$

Applying Jensen's inequality and $(a + b)^2 \le 2a^2 + 2b^2$ to Eq. (S.24) gives

$$\left\|\Delta\mathbf{p}_i^{+}(\mathbf{S}; \mathbf{S}^{\text{ref}})\right\|_{\boldsymbol{\beta}_i}^2 \le 2c_{\varepsilon_W}^2(1 - K^{-1}) \left( \|\boldsymbol{\delta}_i\|_2^2 + \frac{1}{d_i} \sum_{j\in\mathcal{N}_i^{\text{out}}} \|\boldsymbol{\delta}_j\|_2^2 \right). \tag{S.25}$$

Summing the neighbor term by the membership-vector index yields

$$\sum_{i\in\mathcal{N}}\frac{1}{d_i}\sum_{j\in\mathcal{N}_i^{\text{out}}}\|\boldsymbol{\delta}_j\|_2^2=\sum_{i\in\mathcal{N}}\left(\sum_{j\in\mathcal{N}_i^{\text{in}}}\frac{1}{d_j}\right)\|\boldsymbol{\delta}_i\|_2^2\le\chi_{\mathbf{A}}\|\mathbf{S}-\mathbf{S}^{\text{ref}}\|_F^2. \tag{S.26}$$

Consequently,

$$C_{\boldsymbol{\Delta}P}^{\boldsymbol{\beta}}(\mathbf{S};\mathbf{S}^{\text{ref}})\le\frac{2c_{\varepsilon_W}^2(1-K^{-1})(1+\chi_{\mathbf{A}})}{N}\|\mathbf{S}-\mathbf{S}^{\text{ref}}\|_F^2. \tag{S.27}$$

Since $0<c_{\varepsilon_W}^2<1$, Eq. (S.27) implies the bound in Proposition 3. The coefficient reduction is strict whenever $\mathbf{S}\neq\mathbf{S}^{\text{ref}}$; when $\mathbf{S}=\mathbf{S}^{\text{ref}}$, all terms vanish. □

### S.2.6 Proof of Proposition 4

Let $N\ge 2$, $d_{\max}=\max_i d_i$, and $\eta=\varepsilon_W/d_{\max}$. For every feasible membership profile and every stage $t$, the weighting rule in the main article gives

$$W_{ii}(\mathbf{S}(t))\ge\varepsilon_W\ge\eta,\qquad W_{ij}(\mathbf{S}(t))\ge\frac{\varepsilon_W}{d_i}\ge\eta\quad\text{for every } j\in\mathcal{N}_i^{\text{out}}. \tag{S.28}$$

Thus, all diagonal entries and all entries supported by the fixed trust network share the positive lower bound $\eta$, independently of the membership sequence.

Set $B=N-1$ and consider any $B$-stage product

$$\mathbf{Q}(t):=\mathbf{W}(\mathbf{S}(t+B-1))\cdots\mathbf{W}(\mathbf{S}(t)),\qquad t\ge 1. \tag{S.29}$$

Strong connectivity implies that, for each ordered pair $(i,j)$, the trust graph contains a directed path from $i$ to $j$ of length at most $B$. The positive diagonal entries can hold the path at a node during unused positions in the product. Consequently, one term in the expansion of $Q_{ij}(t)$ is a product of $B$ entries, each at least $\eta$, and hence

$$Q_{ij}(t)\ge\eta^B\qquad\text{for all } i,j\in\mathcal{N}\text{ and } t\ge 1. \tag{S.30}$$

For a scalar vector $\mathbf{x}$, write $R(\mathbf{x}):=\max_i x_i-\min_i x_i$. Any row-stochastic matrix maps every component into the convex hull of the components of $\mathbf{x}$ and therefore cannot increase $R$. Moreover, because every entry of $\mathbf{Q}(t)$ is at least $a:=\eta^B$, it can be decomposed as

$$\mathbf{Q}(t)=a\mathbf{1}\mathbf{1}^{\top}+(1-Na)\widetilde{\mathbf{Q}}(t), \tag{S.31}$$

where $\widetilde{\mathbf{Q}}(t)$ is row stochastic when $1-Na>0$; the case $1-Na=0$ gives identical rows directly. The first term produces the same scalar in every row, so

$$R(\mathbf{Q}(t)\mathbf{x})\le\theta R(\mathbf{x}),\qquad\theta:=1-N\eta^B\in[0,1). \tag{S.32}$$

Apply this contraction independently to each alternative. If $R_\ell(t):=\max_i p_{i\ell}(t)-\min_i p_{i\ell}(t)$, then

$$R_\ell(t)\le\theta^{\lfloor t/B\rfloor}R_\ell(0)\longrightarrow 0. \tag{S.33}$$

It follows that $|p_{i\ell}(t)-p_{j\ell}(t)|\to 0$ for all $i,j$ and $\ell$, and therefore $\|\mathbf{p}_i(t)-\mathbf{p}_j(t)\|_2\to 0$. Since every $p_{i\ell}(t)$ and the weighted collective value $\bar{p}_\ell(t)$ lie between the corresponding population minimum and maximum,

$$0\le 1-GCL(t)=\frac{1}{N}\sum_i\sum_{\ell=1}^{L}\beta_{i\ell}|p_{i\ell}(t)-\bar{p}_\ell(t)|\le\sum_{\ell=1}^{L}R_\ell(t)\longrightarrow 0. \tag{S.34}$$

Thus, $GCL(t)\to 1$. For every $\mu\in(0,1)$, convergence supplies a finite $T_\mu$ such that $GCL(T_\mu)\ge\mu$. If the CRP has not stopped earlier, its stopping rule is therefore met by that stage. □

**Phase S.1** Topology-driven local overlapping-community initialization

**Require:** fixed directed adjacency $\mathbf{A}$, edge set $\mathcal{E}$, and resolution $\gamma_{\mathrm{dir}}$;
**Ensure:** crisp-overlapping communities $\{\mathcal{B}_1, \ldots, \mathcal{B}_K\}$, incidence matrix $\mathbf{Z}$, and $\mathbf{S}(0)$;
Construct $\mathcal{D}_i$ for every $i \in \mathcal{N}$ and set $\mathcal{U} \leftarrow \mathcal{N}$;
Initialize an uncovered-count priority queue for seed selection;
**while** $\mathcal{U} \neq \varnothing$ **do**
    Select $i_k \leftarrow \arg\max_{i \in \mathcal{U}}(|\mathcal{D}_i \cap \mathcal{U}|, |\mathcal{D}_i|, -i)$;
    Set $\mathcal{B}_k \leftarrow \{i_k\}$ and initialize its boundary max-heap with cached gains $\widehat{\Delta Q}_{\mathrm{loc}}$;
    **while** the largest cached boundary gain is positive **do**
        Extract all valid boundary nodes with positive cached gain into $\mathcal{V}_k$;
        Add $\mathcal{V}_k$ to $\mathcal{B}_k$ and update $m_{\mathrm{in}}(\mathcal{B}_k)$, $d_{\mathrm{out}}(\mathcal{B}_k)$, and $d_{\mathrm{in}}(\mathcal{B}_k)$;
        Refresh only boundary entries incident to $\mathcal{V}_k$ and insert newly exposed boundary nodes;
    **end while**
    Append $\mathcal{B}_k$ to the community list and set $\mathcal{U} \leftarrow \mathcal{U} \setminus \mathcal{B}_k$;
    Update uncovered-count priorities for seeds whose $\mathcal{D}_i$ intersects $\mathcal{B}_k$;
**end while**
Set $Z_{ik} \leftarrow \mathbf{1}\{i \in \mathcal{B}_k\}$ and $s_{ik}(0) \leftarrow Z_{ik} / \sum_{r=1}^{K} Z_{ir}$;
**return** $\{\mathcal{B}_k\}_{k=1}^{K}$, $\mathbf{Z}$, and $\mathbf{S}(0)$;

## S.3 Supplementary algorithm implementation details

This section specifies the two topology-only phases, the compatible-class batched response sweep, and the deterministic simplex-QP candidate-and-acceptance rule invoked by Algorithm 1, and derives the complexity and termination properties of the complete process. The phases are computed once from the fixed directed topology; the QP rule is applied during each compatible-class response sweep.

### S.3.1 Topology-driven membership initialization

For each DM $i$, let $\mathcal{D}_i = \{i\} \cup \mathcal{N}_i^{\mathrm{in}} \cup \mathcal{N}_i^{\mathrm{out}}$ be its closed directed incident neighborhood. Phase S.1 repeatedly selects a seed by uncovered local reach and expands its community with boundary nodes having positive cached directed-modularity gain. Batched admissions permit overlapping communities while avoiding a full-network gain recomputation after every admitted node. Row-normalizing the final crisp incidence matrix produces the initial membership profile $\mathbf{S}(0)$ used in the main algorithm. Phase S.1 returns both the structural cover and its simplex-valued representation. Its cached-gain updates implement the local-stale expansion described in the main text without changing the directed local-modularity criterion.

### S.3.2 Impact-set response scheduling

SCMCF uses the topology-derived impact set $\mathcal{F}_i = \{i\} \cup \mathcal{N}_i^{\mathrm{in}}$. Phase S.2 forms a compatible partition without materializing the response-conflict graph. Its deterministic ordering prioritizes larger impact sets and then larger out-degrees, with node index resolving remaining ties. First-fit assignment places each DM in the earliest class whose occupied impact set is disjoint from its own. Every class returned by Phase S.2 satisfies Definition 3. The partition is fixed across feedback stages and supplies the class order used by Procedure S.1.

**Phase S.2** Impact-set ordered first-fit response scheduling

**Require:** fixed directed adjacency $\mathbf{A}$;
**Ensure:** response classes $\{\mathcal{C}_1, \ldots, \mathcal{C}_\Gamma\}$;
Construct $\mathcal{F}_i = \{i\} \cup \mathcal{N}_i^{\text{in}}$ for every $i \in \mathcal{N}$;
Order DMs as $\boldsymbol{\pi}$ by $(-|\mathcal{F}_i|, -d_i^{\text{out}}, i)$;
Initialize empty classes and occupied impact sets $\mathcal{O}_g$;
**for** $\ell = 1, \ldots, N$ **do**
    Set $g \leftarrow \min\{r : \mathcal{F}_{\pi_\ell} \cap \mathcal{O}_r = \varnothing\}$, creating a new class if no feasible $r$ exists;
    Add $\pi_\ell$ to $\mathcal{C}_g$ and update $\mathcal{O}_g \leftarrow \mathcal{O}_g \cup \mathcal{F}_{\pi_\ell}$;
**end for**
**return** $\{\mathcal{C}_1, \ldots, \mathcal{C}_\Gamma\}$;

**Procedure S.1** Compatible-class batched response sweep

**Require:** current preference profile $\mathbf{P}$, incumbent membership profile $\mathbf{S}^{\text{ref}}$, retained-membership reference $\bar{\mathbf{p}}^{\text{ref}}$, weights $\{\beta_{i\ell}\}$, compatible classes $\{\mathcal{C}_1, \ldots, \mathcal{C}_\Gamma\}$, and $\alpha$;
**Ensure:** updated membership profile $\mathbf{S}^{(\Gamma)}$;
Set $\mathbf{S}^{(0)} \leftarrow \mathbf{S}^{\text{ref}}$ and initialize the quadratic contributions;
**for** $g = 1, \ldots, \Gamma$ **do**
    Freeze $\mathbf{S}^{(g-1)}$ as the pre-activation profile for class $\mathcal{C}_g$;
    Form the class QPs from the affected prospective preferences and their $\alpha$-weighted targets;
    **for** each $i \in \mathcal{C}_g$ **do**
        Apply Subprocedure S.1 to the QP with frozen opponents $\mathbf{S}_{-i}^{(g-1)}$, obtaining $\widehat{\mathbf{s}}_i$;
    **end for**
    Commit $\{\widehat{\mathbf{s}}_i : i \in \mathcal{C}_g\}$ simultaneously and update the affected quadratic contributions;
**end for**
**return** $\mathbf{S}^{(\Gamma)}$;

### S.3.3 Deterministic simplex-QP candidate-and-acceptance subprocedure

For a fixed preference profile, retained reference, and pre-activation membership profile, DM $i$'s response problem has the simplex-constrained convex quadratic form

$$\min_{\mathbf{s}_i \in \Delta_K} \frac{1}{2}\mathbf{s}_i^\top \mathbf{H}_i \mathbf{s}_i + \mathbf{q}_i^\top \mathbf{s}_i. \tag{S.35}$$

Section S.2.2 establishes $\mathbf{H}_i \succeq \mathbf{0}$, so the simplex problem has a nonempty compact minimizer set. This set is the best-response set analyzed in the main theory.

The numerical rule starts from the incumbent membership. When $\mathbf{H}_i = \mathbf{0}$, Eq. (S.35) is linear, and the rule projects the incumbent onto the optimal simplex face supported by $\arg\min_k q_{ik}$. For $\mathbf{H}_i \neq \mathbf{0}$, it applies projected FISTA with a fixed valid Lipschitz constant. One iteration is

$$\mathbf{x}_i^{(r)} = \Pi_{\Delta_K}\left(\mathbf{y}_i^{(r-1)} - L_i^{-1}(\mathbf{H}_i \mathbf{y}_i^{(r-1)} + \mathbf{q}_i)\right), \quad \mathbf{y}_i^{(r)} = \mathbf{x}_i^{(r)} + \frac{\tau_{r-1} - 1}{\tau_r}\left(\mathbf{x}_i^{(r)} - \mathbf{x}_i^{(r-1)}\right), \tag{S.36}$$

where $L_i \geq \lambda_{\max}(\mathbf{H}_i)$, $\tau_0 = 1$, and $\tau_r = (1 + \sqrt{1 + 4\tau_{r-1}^2})/2$. Subprocedure S.1 gives the complete deterministic simplex-QP candidate-and-acceptance rule. It returns a feasible simplex candidate $\widetilde{\mathbf{s}}_i$ and commits that candidate when the frozen local cost decreases by at least $\varepsilon_{\text{QP}}$; otherwise, it retains the incumbent membership. This candidate map provides the numerical realization of

**Subprocedure S.1** Deterministic simplex-QP candidate-and-acceptance rule

**Require:** $\mathbf{H}_i \succeq \mathbf{0}$, $\mathbf{q}_i$, incumbent $\mathbf{s}_i^0 \in \Delta_K$, fixed opponents $\mathbf{S}_{-i}$, local cost $J_i$, QP tolerance $\varepsilon_{\mathrm{QP}}$, and iteration cap $I_{\max}$;

**Ensure:** committed candidate $\widehat{\mathbf{s}}_i$;

**if** $\mathbf{H}_i = \mathbf{0}$ **then**
  $\mathcal{I}_i^\star \leftarrow \arg\min_{k\in\mathcal{K}} q_{ik}$;
  $\mathcal{F}_i^\star \leftarrow \{\mathbf{s} \in \Delta_K : s_k = 0 \text{ for } k \notin \mathcal{I}_i^\star\}$;
  $\widetilde{\mathbf{s}}_i \leftarrow \Pi_{\mathcal{F}_i^\star}(\mathbf{s}_i^0)$;
**else**
  Choose fixed $L_i \geq \lambda_{\max}(\mathbf{H}_i)$; set $\mathbf{x}_i^{(0)} \leftarrow \mathbf{y}_i^{(0)} \leftarrow \mathbf{s}_i^0$, $\tau_0 \leftarrow 1$, and $r_{\mathrm{stop}} \leftarrow I_{\max}$;
  **for** $r = 1, \ldots, I_{\max}$ **do**
    $\mathbf{x}_i^{(r)} \leftarrow \Pi_{\Delta_K}\Big(\mathbf{y}_i^{(r-1)} - L_i^{-1}(\mathbf{H}_i\mathbf{y}_i^{(r-1)} + \mathbf{q}_i)\Big)$;
    $\tau_r \leftarrow (1 + \sqrt{1 + 4\tau_{r-1}^2})/2$;
    $\mathbf{y}_i^{(r)} \leftarrow \mathbf{x}_i^{(r)} + \frac{\tau_{r-1}-1}{\tau_r}(\mathbf{x}_i^{(r)} - \mathbf{x}_i^{(r-1)})$;
    **if** $\|\mathbf{x}_i^{(r)} - \mathbf{x}_i^{(r-1)}\|_\infty \leq \varepsilon_{\mathrm{QP}}$ **then**
      $r_{\mathrm{stop}} \leftarrow r$; **break**;
    **end if**
  **end for**
  $\widetilde{\mathbf{s}}_i \leftarrow \mathbf{x}_i^{(r_{\mathrm{stop}})}$;
**end if**
**if** $J_i(\widetilde{\mathbf{s}}_i, \mathbf{S}_{-i}) \leq J_i(\mathbf{s}_i^0, \mathbf{S}_{-i}) - \varepsilon_{\mathrm{QP}}$ **then**
  $\widehat{\mathbf{s}}_i \leftarrow \widetilde{\mathbf{s}}_i$;
**else**
  $\widehat{\mathbf{s}}_i \leftarrow \mathbf{s}_i^0$;
**end if**
**return** $\widehat{\mathbf{s}}_i$;

the local response specified by the QP. Because compatible DMs have invariant local QPs under one another's updates, applying the same candidate map preserves within-class serial–batch reproducibility. The response coefficients can be maintained without recomputing every trusted-source sum after each class. Let $m_{hj}(\mathbf{S}) := \mathbf{s}_h^\top \mathbf{s}_j$ and let $\mathbf{a}_h$ be the membership-independent term in Eq. (S.11). Define $\mathbf{c}_h^{\mathrm{full}}(\mathbf{S}) := \mathbf{a}_h + (1 - 2\varepsilon_W) d_h^{-1} \sum_{j\in\mathcal{N}_h^{\mathrm{out}}} m_{hj}(\mathbf{S})(\mathbf{p}_j - \mathbf{p}_h)$. If $i \in \mathcal{N}_h^{\mathrm{out}}$, subtracting the $i$-term gives $\mathbf{c}_{h,-i}$. After committing class $g$, these quantities are updated by $\mathbf{c}_h^{\mathrm{full},(g)} = \mathbf{c}_h^{\mathrm{full},(g-1)} + (1 - 2\varepsilon_W) d_h^{-1} \sum_{j\in\mathcal{N}_h^{\mathrm{out}}} \Delta m_{hj}^{(g)} (\mathbf{p}_j - \mathbf{p}_h)$. This changes only coefficient maintenance, not the local QPs represented.

Table S.1: Threshold-attainment endpoints across nine network–preference profiles.

**FilmTrust**

| | *FilmTrust-10* | | | | *FilmTrust-30* | | | | *FilmTrust-50* | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mechanism | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $T_\mu$ | $CI_{\min}$ | TPA | CPA |
| SNDG [1] | 5 | 0.7536949 | 0.1640269 | 0.1790443 | 4 | 0.7676454 | 0.1609090 | 0.1711416 | 4 | 0.7692501 | 0.1660622 | 0.1756310 |
| TEDG [2] | 3 | 0.6307498 | 0.1831787 | 0.3635607 | 3 | 0.7023910 | 0.1852494 | 0.3491323 | 3 | 0.7041296 | 0.1911005 | 0.3503806 |
| DGBC [3] | 3 | 0.6271766 | 0.1822098 | 0.3613448 | 3 | 0.7013226 | 0.1850127 | 0.3489833 | 3 | 0.7040716 | 0.1916427 | 0.3496994 |
| DTRF [4] | **2** | 0.7451293 | 0.1746076 | 0.2776208 | **2** | 0.7529661 | 0.1795978 | 0.2644774 | **2** | 0.7588854 | 0.1842936 | 0.2571174 |
| DTLC [5] | 5 | 0.7816350 | 0.1757552 | 0.2263617 | 5 | 0.7835106 | 0.1772693 | 0.2230305 | 5 | 0.7870962 | 0.1815302 | 0.2238463 |
| OCRF [6] | 4 | 0.2519673 | 0.1609976 | 0.2014237 | 5 | 0.6144543 | 0.1694490 | 0.2210268 | 4 | 0.6308666 | 0.1767742 | 0.2250654 |
| DCRTF [7] | 3 | 0.6956195 | 0.1761486 | 0.3269839 | 3 | 0.8192211 | 0.1801750 | 0.3196366 | **2** | 0.7859034 | 0.1606776 | 0.2873987 |
| OBCF [8] | 3 | 0.5864296 | 0.1611762 | 0.1695854 | 4 | 0.7014344 | 0.1748728 | 0.1858265 | 4 | 0.7050042 | 0.1784742 | 0.1876939 |
| OCCF [9] | **2** | 0.3025599 | 0.1642890 | 0.1828797 | **2** | 0.2661319 | 0.1688364 | 0.1891813 | **2** | 0.2648807 | 0.1759794 | 0.1964324 |
| NGPF [10] | **2** | **0.8182483** | **0.1526424** | 0.2915209 | **2** | **0.8199199** | **0.1561536** | 0.2836299 | **2** | 0.8200987 | **0.1606156** | 0.2862436 |
| SCMCF | 4 | 0.7726567 | 0.1554582 | **0.1626469** | 4 | 0.7838209 | 0.1573543 | **0.1641027** | 4 | **0.8599562** | 0.1649947 | **0.1712214** |

**Ciao**

| | *Ciao-10* | | | | *Ciao-30* | | | | *Ciao-50* | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mechanism | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $T_\mu$ | $CI_{\min}$ | TPA | CPA |
| SNDG [1] | 3 | 0.5830910 | 0.2115864 | 0.2138610 | 3 | 0.5964792 | 0.2115042 | 0.2129640 | 3 | 0.5946463 | 0.2020669 | 0.2032513 |
| TEDG [2] | 2 | 0.6785423 | 0.2345116 | 0.2975281 | **1** | 0.2080551 | 0.2490969 | 0.2490969 | **1** | 0.2110446 | 0.2377531 | 0.2377531 |
| DGBC [3] | 2 | 0.2074377 | 0.2344831 | 0.2971991 | **1** | 0.2080551 | 0.2490969 | 0.2490969 | **1** | 0.2110446 | 0.2377531 | 0.2377531 |
| DTRF [4] | 2 | 0.6517151 | 0.2456942 | 0.2740256 | **1** | 0.2054228 | 0.2270926 | 0.2270926 | **1** | 0.2129832 | 0.2177749 | 0.2177749 |
| DTLC [5] | 5 | 0.5333854 | 0.2390561 | 0.2591372 | 5 | 0.6169724 | 0.2351278 | 0.2498779 | 4 | 0.5161394 | 0.2112807 | 0.2219871 |
| OCRF [6] | 9 | 0.1033502 | 0.2106062 | 0.2419909 | 7 | 0.1189601 | 0.2154867 | 0.2435527 | 6 | 0.1054374 | 0.2052092 | 0.2303675 |
| DCRTF [7] | 2 | 0.7109920 | 0.2341199 | 0.2837407 | **1** | 0.2055610 | 0.2469890 | 0.2469890 | **1** | 0.2095206 | 0.2367582 | 0.2367582 |
| OBCF [8] | 4 | 0.5425970 | 0.2284523 | 0.2372460 | 4 | 0.5175047 | 0.2233071 | 0.2288202 | 4 | 0.5083732 | 0.2129093 | 0.2171789 |
| OCCF [9] | **1** | 0.7016267 | 0.2397380 | 0.2397380 | **1** | 0.6939750 | 0.2418005 | 0.2418005 | **1** | 0.4701590 | 0.2316865 | 0.2316865 |
| NGPF [10] | 2 | **0.7135234** | 0.2355364 | 0.2920384 | **1** | 0.2471488 | 0.2484940 | 0.2484940 | **1** | 0.2501682 | 0.2373855 | 0.2373855 |
| SCMCF | 3 | 0.6699902 | **0.2074065** | **0.2086956** | 3 | **0.7182319** | **0.2088192** | **0.2097658** | 3 | **0.7790617** | **0.2003307** | **0.2011870** |

**Deezer**

| | *Deezer-RO* | | | | *Deezer-HU* | | | | *Deezer-HR* | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mechanism | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $T_\mu$ | $CI_{\min}$ | TPA | CPA |
| SNDG [1] | 2 | 0.6902933 | 0.0665556 | 0.0665556 | 2 | 0.7222075 | 0.0658158 | 0.0658158 | 2 | 0.7698339 | 0.0707936 | 0.0707936 |
| TEDG [2] | 2 | 0.7230529 | 0.0775740 | 0.1520702 | **1** | 0.6345603 | 0.0912832 | 0.0912832 | **1** | 0.6554497 | 0.0965206 | 0.0965206 |
| DGBC [3] | 2 | 0.7227566 | 0.0775620 | 0.1521086 | **1** | 0.6345603 | 0.0912832 | 0.0912832 | **1** | 0.6554497 | 0.0965206 | 0.0965206 |
| DTRF [4] | 2 | 0.7368145 | 0.0683580 | 0.0696490 | 2 | **0.7939372** | 0.0662186 | 0.0672751 | 2 | 0.7348461 | 0.0677063 | 0.0681349 |
| DTLC [5] | 5 | **0.7932255** | 0.1174283 | 0.1179933 | 4 | 0.7722382 | 0.1002424 | 0.1004269 | 4 | **0.8178228** | 0.1109168 | 0.1110827 |
| OCRF [6] | 2 | 0.6982473 | **0.0573414** | 0.0698255 | 2 | 0.7154214 | 0.0560987 | 0.0688530 | 2 | 0.6978568 | **0.0586693** | 0.0714823 |
| DCRTF [7] | 2 | 0.5655770 | 0.0648230 | 0.1251811 | 2 | 0.6211616 | 0.0741891 | 0.1273528 | **1** | 0.6522532 | 0.0913695 | 0.0913695 |
| OBCF [8] | 4 | 0.7479794 | 0.1145371 | 0.1145438 | 3 | 0.7540451 | 0.0888175 | 0.0888178 | 3 | 0.7645956 | 0.0940953 | 0.0940955 |
| OCCF [9] | **1** | 0.4868347 | 0.0689903 | 0.0689903 | **1** | 0.7165134 | 0.0713780 | 0.0713780 | **1** | 0.7115651 | 0.0806067 | 0.0806067 |
| NGPF [10] | 2 | 0.7513995 | 0.0790104 | 0.1471409 | **1** | 0.6521555 | 0.0905222 | 0.0905222 | **1** | 0.6720203 | 0.0956000 | 0.0956000 |
| SCMCF | 3 | 0.7157436 | 0.0645391 | **0.0650706** | 2 | 0.6059074 | **0.0533983** | **0.0535490** | 2 | 0.7928361 | 0.0618545 | **0.0620096** |

Lower values are better for $T_\mu$, TPA, and CPA; higher values are better for $CI_{\min}$. Bold marks every best value within each profile. TPA and CPA are per reported DM–alternative cell. Values are shown to seven decimal places; comparisons use stored values with exact equality for $T_\mu$ and an absolute tolerance of $10^{-12}$ for other metrics.

Table S.2: SCMCF ablation endpoints across nine network–preference profiles.

| **FilmTrust** | *FilmTrust-10* | | | | | *FilmTrust-30* | | | | | *FilmTrust-50* | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Variant | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $\mathrm{TR}_v$ | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $\mathrm{TR}_v$ | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $\mathrm{TR}_v$ |
| SCMCF-Bal | 5 | 0.7935 | 0.1569 | 0.1657 | 1.1400 | 5 | **0.8559** | 0.1627 | 0.1724 | 1.1665 | 5 | 0.8263 | 0.1670 | 0.1753 | 1.1118 |
| SCMCF-Ser | 4 | 0.7727 | 0.1555 | 0.1626 | 1.9669 | 4 | 0.7838 | 0.1574 | 0.1641 | 1.9898 | 4 | 0.8600 | 0.1650 | 0.1712 | 1.9392 |
| SCMCF-Static | 7 | 0.7503 | 0.1629 | 0.2516 | **0.0213** | 8 | 0.7959 | 0.1623 | 0.2274 | **0.0289** | 6 | 0.8283 | 0.1632 | 0.2325 | **0.0424** |
| SCMCF-FC | 4 | 0.7718 | **0.1537** | **0.1598** | 1.0300 | 5 | 0.8092 | 0.1624 | 0.1708 | 1.3527 | 5 | 0.8458 | 0.1708 | 0.1793 | 1.2264 |
| SCMCF-Own | 5 | 0.7307 | 0.1550 | 0.1650 | 0.9075 | 4 | 0.7976 | **0.1541** | **0.1616** | 0.8204 | 4 | 0.7218 | **0.1614** | **0.1679** | 0.7704 |
| SCMCF-NoMA | **3** | 0.7138 | 0.1616 | 0.1881 | 0.4814 | **3** | 0.7944 | 0.1618 | 0.1883 | 0.4790 | **3** | 0.8589 | 0.1691 | 0.1968 | 0.4360 |
| SCMCF-NE | 4 | **0.8023** | 0.1582 | 0.1656 | 14.9718 | 4 | 0.8308 | 0.1595 | 0.1665 | 27.5372 | 4 | **0.8921** | 0.1674 | 0.1746 | 15.1430 |
| SCMCF | 4 | 0.7727 | 0.1555 | 0.1626 | 1.0000 | 4 | 0.7838 | 0.1574 | 0.1641 | 1.0000 | 4 | 0.8600 | 0.1650 | 0.1712 | 1.0000 |
| **Ciao** | *Ciao-10* | | | | | *Ciao-30* | | | | | *Ciao-50* | | | | |
| Variant | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $\mathrm{TR}_v$ | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $\mathrm{TR}_v$ | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $\mathrm{TR}_v$ |
| SCMCF-Bal | 4 | **0.8194** | 0.2102 | 0.2120 | 1.2402 | 4 | **0.8198** | 0.2109 | 0.2122 | 1.2486 | 4 | **0.7902** | 0.2011 | 0.2023 | 1.2930 |
| SCMCF-Ser | 3 | 0.6700 | 0.2074 | **0.2087** | 1.5707 | 3 | 0.7182 | 0.2088 | 0.2098 | 1.5775 | 3 | 0.7791 | 0.2003 | 0.2012 | 1.6202 |
| SCMCF-Static | 4 | 0.1085 | **0.2034** | 0.2090 | **0.0615** | 4 | 0.5156 | **0.1994** | **0.2058** | **0.0711** | 4 | 0.5138 | **0.1908** | **0.1955** | **0.0922** |
| SCMCF-FC | 4 | 0.7262 | 0.2225 | 0.2244 | 1.2953 | 4 | 0.6815 | 0.2227 | 0.2242 | 1.3059 | 4 | 0.7175 | 0.2135 | 0.2147 | 1.2950 |
| SCMCF-Own | 3 | 0.6508 | 0.2101 | 0.2116 | 0.7521 | 3 | 0.7054 | 0.2134 | 0.2147 | 0.4149 | 3 | 0.6725 | 0.2017 | 0.2028 | 0.3854 |
| SCMCF-NoMA | **2** | 0.5234 | 0.2064 | 0.2114 | 0.7938 | **2** | 0.6395 | 0.2130 | 0.2182 | 0.5418 | **2** | 0.7000 | 0.2070 | 0.2128 | 0.5448 |
| SCMCF-NE | 3 | 0.7443 | 0.2092 | 0.2101 | 52.6486 | 3 | 0.7697 | 0.2100 | 0.2107 | 46.5443 | 3 | 0.7817 | 0.2019 | 0.2026 | 52.1494 |
| SCMCF | 3 | 0.6700 | 0.2074 | **0.2087** | 1.0000 | 3 | 0.7182 | 0.2088 | 0.2098 | 1.0000 | 3 | 0.7791 | 0.2003 | 0.2012 | 1.0000 |
| **Deezer** | *Deezer-RO* | | | | | *Deezer-HU* | | | | | *Deezer-HR* | | | | |
| Variant | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $\mathrm{TR}_v$ | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $\mathrm{TR}_v$ | $T_\mu$ | $CI_{\min}$ | TPA | CPA | $\mathrm{TR}_v$ |
| SCMCF-Bal | 4 | 0.6805 | 0.0676 | 0.0686 | 1.3700 | 3 | **0.7886** | 0.0572 | 0.0576 | 1.0092 | 3 | 0.7890 | 0.0636 | 0.0641 | 1.0089 |
| SCMCF-Ser | 3 | 0.7157 | 0.0645 | 0.0651 | 7.4430 | **2** | 0.6059 | 0.0534 | 0.0535 | 4.8730 | **2** | **0.7928** | 0.0619 | 0.0620 | 2.5887 |
| SCMCF-Static | 25 | **0.7415** | 0.0594 | 0.0608 | 0.7103 | 12 | 0.6706 | 0.0521 | 0.0524 | 0.6010 | 6 | 0.7528 | **0.0574** | **0.0575** | 0.3126 |
| SCMCF-FC | 3 | 0.7157 | 0.0645 | 0.0651 | 1.0050 | **2** | 0.6059 | 0.0534 | 0.0535 | 0.9960 | **2** | **0.7928** | 0.0619 | 0.0620 | 1.0080 |
| SCMCF-Own | 3 | 0.6376 | 0.0591 | 0.0600 | **0.6078** | **2** | 0.6060 | **0.0510** | **0.0514** | **0.3531** | **2** | 0.7054 | 0.0614 | 0.0619 | **0.2297** |
| SCMCF-NoMA | 3 | 0.7383 | 0.0747 | 0.0821 | 0.9670 | **2** | 0.6062 | 0.0647 | 0.0677 | 0.9891 | **2** | 0.7460 | 0.0747 | 0.0778 | 1.0172 |
| SCMCF-NE | **2** | 0.6989 | **0.0573** | **0.0573** | 64.6168 | **2** | 0.7676 | 0.0581 | 0.0582 | 80.1318 | **2** | 0.7921 | 0.0653 | 0.0653 | 111.9327 |
| SCMCF | 3 | 0.7157 | 0.0645 | 0.0651 | 1.0000 | **2** | 0.6059 | 0.0534 | 0.0535 | 1.0000 | **2** | **0.7928** | 0.0619 | 0.0620 | 1.0000 |

Lower values are better for $T_\mu$, TPA, CPA, and $\mathrm{TR}_v$; higher values are better for $CI_{\min}$. Bold marks every best value within each profile. The paired runtime ratio is $\mathrm{TR}_v := \tau_v/\tau_{\mathrm{SCMCF}}$, where $\tau_v$ is the runtime of variant $v$; $\mathrm{TR}_v > 1$ favors SCMCF and $\mathrm{TR}_v < 1$ favors $v$. Runtimes are single measurements of variant execution in the shared environment. TPA and CPA are per reported DM–alternative cell. Values are shown to four decimal places; comparisons use stored values with exact equality for $T_\mu$ and an absolute tolerance of $10^{-12}$ for other metrics.

Table S.3: SCMCF endpoints under controlled observation retention.

| Network-preference profile | Retention (%) | Attained/n | $T_\mu$ | $CI_{\min}$ | TPA | CPA |
|---|---|---|---|---|---|---|
| Deezer-RO | 100 | 1/1 | 3 | 0.7157 | 0.0645 | 0.0651 |
| | 80 | 20/20 | 3 [0] | 0.7425 [0.0108] | 0.0659 [0.0001] | 0.0665 [0.0001] |
| | 60 | 20/20 | 3 [0] | 0.7562 [0.0187] | 0.0673 [0.0002] | 0.0678 [0.0002] |
| | 40 | 20/20 | 2 [0] | 0.6018 [0.0341] | 0.0517 [0.0002] | 0.0518 [0.0002] |
| | 20 | 20/20 | 2 [0] | 0.5212 [0.0569] | 0.0532 [0.0003] | 0.0532 [0.0003] |
| Deezer-HU | 100 | 1/1 | 2 | 0.6059 | 0.0534 | 0.0535 |
| | 80 | 20/20 | 2 [0] | 0.6121 [0.0379] | 0.0545 [0.0001] | 0.0547 [0.0001] |
| | 60 | 20/20 | 2 [0] | 0.6276 [0.0332] | 0.0556 [0.0001] | 0.0557 [0.0001] |
| | 40 | 20/20 | 2 [0] | 0.6328 [0.0520] | 0.0566 [0.0001] | 0.0567 [0.0001] |
| | 20 | 20/20 | 2 [0] | 0.6663 [0.1129] | 0.0575 [0.0003] | 0.0575 [0.0003] |
| Deezer-HR | 100 | 1/1 | 2 | 0.7928 | 0.0619 | 0.0620 |
| | 80 | 20/20 | 2 [0] | 0.7836 [0.0162] | 0.0628 [0.0001] | 0.0629 [0.0001] |
| | 60 | 20/20 | 2 [0] | 0.7895 [0.0314] | 0.0636 [0.0001] | 0.0637 [0.0001] |
| | 40 | 20/20 | 2 [0] | 0.7799 [0.0128] | 0.0643 [0.0002] | 0.0644 [0.0002] |
| | 20 | 20/20 | 2 [0] | 0.7431 [0.0185] | 0.0649 [0.0003] | 0.0649 [0.0003] |

At 100% retention, each row is the deterministic complete-profile reference. Other rows report the median [IQR] over 20 mask seeds. All endpoints attain $GCL \geq 0.95$ within 100 rounds. TPA and CPA are normalized by the number of current observed cells.

### S.3.4 Complexity and termination of the complete process

The computational cost has two components: topology-only preprocessing and repeated feedback stages. Let $H_Q$ be the total number of cached boundary-heap operations performed while constructing the overlapping cover. Phase S.1 costs $O(M + H_Q \log N)$. With hash-set impact representations, Phase S.2 costs at most $O(\Gamma(N + M))$ because each DM may test up to $\Gamma$ occupied impact sets. These two topology costs are paid once and determine the membership dimension $K$ and response-class span $\Gamma$ for all subsequent stages.

Within one feedback stage, forecast construction and QP assembly contribute $O(M(LK + L + K^2) + NLK^2)$. Since $I_i \leq I_{\max}$, the projected-FISTA solves contribute at most $O(NI_{\max}(K^2 + K \log K))$; eigenvalue-based step sizes add $O(NK^3)$, or $O(NK^2)$ with a row-sum Lipschitz bound. For $L, K \geq 1$, define the compact eigenvalue-based bound by $\mathcal{C}_{\text{stage}} := O(M(LK + K^2) + N(LK^2 + K^3 + I_{\max}(K^2 + K \log K)))$. The largest component determines per-stage cost: $MLK$ and $NLK^2$ represent network–preference processing, $NK^3$ represents eigenvalue-based step-size construction, and $NI_{\max}(K^2 + K \log K)$ represents the QP iterations. The bound exposes how network size, preference dimension, community dimension, and iteration cap affect cost.

The compatible-class schedule changes the synchronization span rather than the number of local QPs, since $\sum_g |\mathcal{C}_g| = N$. The community dimension $K$, which can change with $\gamma_{\text{dir}}$, controls coefficient assembly and response cost; $\Gamma$ determines the number of sequential class updates. Over $T$ feedback stages, the repeated work is $T$ times the per-stage bound in addition to the one-time topology cost. Algorithm 1 stops when $GCL(\mathbf{P}(t)) \geq \mu$ or $t = T_{\max}$, and records the corresponding threshold-attainment or stage-limit outcome after the completed class sweep. Proposition 4 supplies finite-threshold attainment for the untruncated calibrated evolution on a strongly connected topology, while Corollary 2 provides the classwise potential structure used by the schedule.

## S.4 Supplementary information for the experiments

This section collects the additional material behind the experiments of the main article. Table S.1 reports the threshold-attainment endpoints of the eleven mechanisms on each of the nine network–preference profiles. Table S.2 reports the threshold-attainment endpoints and paired runtime ratios of seven controlled variants and the complete SCMCF configuration across the nine network–preference profiles. Table S.3 summarizes the controlled-incompleteness analysis: the complete profiles provide one deterministic reference per network, and each incomplete retention level summarizes 20 nested-mask seeds under the same SCMCF configuration, quantifying mask-induced variability on each fixed network.

### S.4.1 Illustration of SCMCF

Figure S.1 trace SCMCF's complete execution on FilmTrust-10 under the default configuration. The initial assessment gives $GCL(\mathbf{P}(0)) = 0.8199 < \mu$, so feedback activates and the threshold is attained at $t = 4$. Panels (a)–(c) show the structural states: Phase S.1 initializes $K = 5$ overlapping communities averaging 2.36 labels per DM, the conflict relation of panel (b) joins 2,873 incompatible pairs and Phase S.2 packs the 211 DMs into $\Gamma = 47$ compatible classes, and the terminal labels recovered from $\mathbf{S}(4)$ at the $1/K$ threshold leave 136 single-label and 75 still-overlapping DMs. Panel (d) shows $\mathbf{P}^\top(t)$ and $\mathbf{S}^\top(t)$ in a fixed DM order, where dimmed cells mark the 40.52% of completed entries that participate in preference evolution but not in assessment. Accepted responses shrink from 186 in the first stage to 136 in the last, every stage reduces the potential under its fixed reference ($\Delta\Psi_1 = 2.8024$ down to $\Delta\Psi_4 = 0.0111$), and the per-stage CPA

increments attenuate from 0.0708 to 0.0167, with the weakest-DM consistency rising from 0.2315 to 0.7727—the controlled-displacement trajectory predicted by Corollary 1 and Proposition 3.

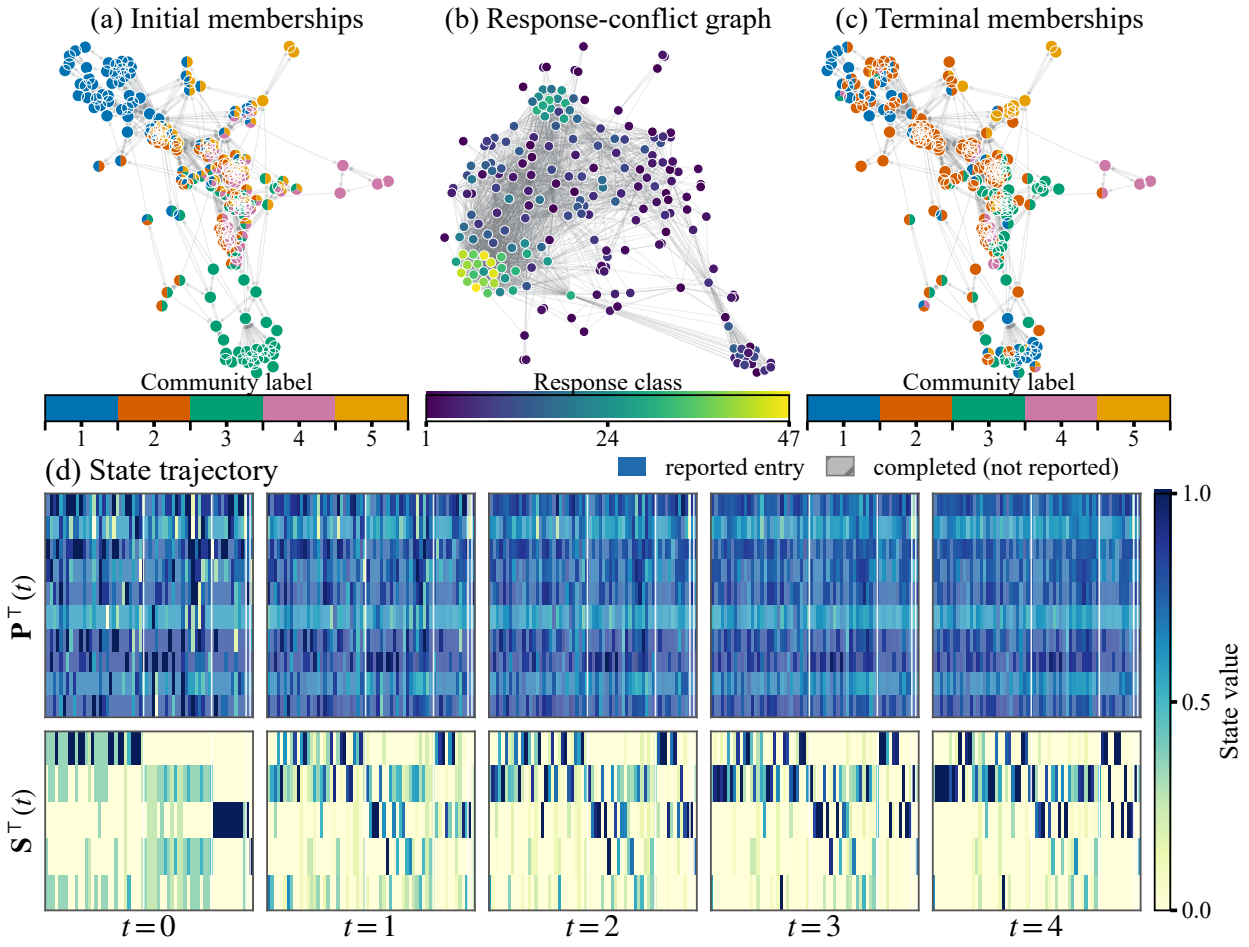



Figure S.1: SCMCF mechanism illustration on FilmTrust-10 under the default configuration. (a) Initial overlapping memberships on the fixed directed trust network, with pie glyphs encoding $\mathbf{S}(0)$. (b) Response-conflict graph colored by compatible class. (c) Terminal memberships with crisp labels recovered from $\mathbf{S}(4)$ at the $1/K$ threshold, sharing the layout and colors of panel (a). (d) $\mathbf{P}^\top(t)$ (upper; dimmed cells are completed, not reported) and $\mathbf{S}^\top(t)$ (lower) for $t = 0, \ldots, 4$ in a fixed DM order on the common $[0, 1]$ scale.

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